

Chapter 2: Continuous-Time Stochastic Optimal Control

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Abstract. *In this chapter, we discuss the continuous-time, continuous-state stochastic optimal control problem. We will focus on the control-affine state space models with quadratic control cost functions, and derive the stochastic Hamilton-Jacobi-Bellman (HJB) equation to characterize the optimal full-state feedback. We will also see how the stochastic HJB equation reduces to a Riccati differential equation in the linear-quadratic-Gaussian (LQG) settings. We will then formulate the nonlinear filtering problem for partially observable systems.*

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2.1 Problem Formulation

Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $W_t \in \mathbb{R}^m$, $t \geq t_0$, denote the standard Brownian motion under P , with $W_{t_0} = 0$. Let $\mathcal{F}_t \subseteq \mathcal{F}$ be an increasing family of σ -algebras such that W_t is a martingale with respect to $\mathcal{F}_t$. Consider the continuous-time dynamical system whose state evolution is modeled by the following n -dimensional process over $t \in [t_0, t_f]$, where t_f is an $\mathcal{F}_t$ -measurable stopping time:

$$dX_t = (f_t(X_t) + g_t(X_t)u_t) dt + \sigma_t(X_t)dW_t, \quad X_{t_0} = x_0 \in \mathbb{R}^n. \quad (2.1)$$

We assume that the functions f_t , g_t , and σ_t are given. We will interpret the SDE (2.1) in the sense of Ito, and thus will require that $f_t(X_t) + g_t(X_t)u_t$ satisfies conditions (i)–(iii) in Section 1.3, and that $\sigma_t(X_t)$ satisfies conditions (i)’–(iii)’ in the same section. Although (2.1) represents a nonlinear state-space model, it is affine in the control input u_t . Such systems are called *control-affine*. This structural assumption simplifies the derivation of the stochastic HJB equation below while maintaining broad applicability. The cost function to be minimized is

$$C(x_0, t_0, u_{[t_0, \infty)}) = \mathbb{E} \left[\int_{t_0}^{t_f} (V_t(X_t) + S_t(X_t)u_t + \frac{1}{2}u_t^\top R_t(X_t)u_t) dt + \psi(X_{t_f}) \right], \quad (2.2)$$

where $S_t(x)$ is a state-dependent row vector and $\psi(\cdot)$ is the terminal cost. We assume that $R_t(x)$ is a positive definite matrix for each t and x . We now formulate the stochastic optimal control

problem as:

$$\min_{u_{[t_0, \infty)}} C(x_0, t_0, u_{[t_0, \infty)}) \quad (2.3a)$$

$$\text{s.t. Dynamics given by (2.1).} \quad (2.3b)$$

Here, u_t is required only to be $\mathcal{F}_t$ -adapted. Since the state x_t is also $\mathcal{F}_t$ -adapted, u_t may depend directly on the realization of $x_{[t_0, t]}$. Hence, a full-state feedback policy is admissible for the problem formulated above.

2.2 Stochastic Hamilton–Jacobi–Bellman (HJB) Equation

The function $C(x_0, t_0, u_{[t_0, \infty)})$ defined by (2.2) represents the *cost-to-go* when the control policy $u_{[t_0, \infty)}$ is applied to the dynamics (2.1) over $[t_0, t_f]$ with initial condition x_0 . More generally, for each $(x, t) \in \mathbb{R}^n \times [t_0, t_f]$ and a fixed control policy $u_{[t, \infty)}$, the cost-to-go function $C(x, t, u_{[t, \infty)})$ can be defined by

$$C(x, t, u_{[t, \infty)}) = \mathbb{E} \left[\int_t^{t_f} (V_\tau(X_\tau) + S_\tau u_\tau + \frac{1}{2} u_\tau^\top R_\tau u_\tau) d\tau + \psi(X_{t_f}) \right], \quad (2.4)$$

assuming the state evolves as

$$dX_\tau = (f_\tau(X_\tau) + \tau(X_\tau)u_\tau(X_\tau)) d\tau + \sigma_\tau(X_\tau)dW_\tau, \quad \tau \in [t, t_f], \quad (2.5)$$

with $X_t = x \in \mathbb{R}^n$. For each (x, t) , the optimal control policy (if it exists) is the one that achieves

$$J(x, t) = \min_{u_{[t, \infty)}} C(x, t, u_{[t, \infty)}), \quad (2.6)$$

where $J : \mathbb{R}^n \times [t_0, t_f] \rightarrow \mathbb{R}$ denotes the *value function* associated with the stochastic optimal control problem (2.3).

The next theorem, known as the *verification theorem* [1], states that the value function is characterized as the solution to a partial differential equation (PDE), commonly known as the *stochastic Hamilton–Jacobi–Bellman (HJB) equation*, defined on $\mathbb{R}^n \times [t_0, t_f]$ with a suitable terminal condition. It also provides the form of the optimal control policy in terms of the gradient of the value function. In what follows, we use the shorthand notation:

$$\partial_t J := \frac{\partial J}{\partial t}, \quad \partial_x J := \frac{\partial J}{\partial x}, \quad \partial_x^2 J := \frac{\partial^2 J}{\partial x^2}. \quad (2.7)$$

Theorem 2.1 Suppose there exists a continuous function $J : \mathbb{R}^n \times [t_0, t_f] \rightarrow \mathbb{R}$ such that

- (a) $J(x, t)$ is continuously differentiable in t and twice continuously differentiable in x on $\mathbb{R}^n \times [t_0, t_f]$;
- (b) $J(x, t)$ satisfies the PDE

$$-\partial_t J = -\frac{1}{2} (g_t^\top \partial_x J + S_t^\top)^\top R_t^{-1} (g_t^\top \partial_x J + S_t^\top) + V_t + f_t^\top \partial_x J + \frac{1}{2} \text{Tr}(\sigma_t \sigma_t^\top \partial_x^2 J) \quad (2.8a)$$

$$\forall (x, t) \in \mathbb{R}^n \times [t_0, t_f], \quad (2.8a)$$

$$J(x, t_f) = \psi(x), \quad x \in \mathbb{R}^n. \quad (2.8b)$$

Then the following hold:

(i) $J(x, t)$ is the value function for problem (2.3), i.e., it satisfies (2.6).

(ii) The optimal control policy is given by

$$u_t(X_t) = -R_t^{-1}(g_t^\top \partial_x J + S_t^\top). \quad (2.9)$$

Proof: For convenience, let us introduce the notation $\mathbb{E}^{x,t}[\cdot] := \mathbb{E}[\cdot | X_t = x]$ to denote the conditional expectation. Let $J(x, t)$ be the function satisfying (a) and (b). By Dynkin's formula for the graph of Ito diffusion (1.89), we have

$$\begin{aligned} & \mathbb{E}^{x,t} [J(X_{t_f}, t_f)] \\ &= J(x, t) + \mathbb{E}^{x,t} \left[\int_t^{t_f} \left\{ \partial_\tau J + (f_\tau + g_\tau u_\tau)^\top (\partial_x J) + \frac{1}{2} \text{Tr} (\sigma_\tau \sigma_\tau^\top \partial_x^2 J) \right\} d\tau \right]. \end{aligned} \quad (2.10)$$

By the boundary condition (2.8b), $J(X_{t_f}, t_f) = \psi(X_{t_f})$. Hence, from (2.10), we obtain

$$J(x, t) = \mathbb{E}^{x,t} \psi(X_{t_f}) - \mathbb{E}^{x,t} \left[\int_t^{t_f} \left\{ \partial_\tau J + (f_\tau + g_\tau u_\tau)^\top (\partial_x J) + \frac{1}{2} \text{Tr} (\sigma_\tau \sigma_\tau^\top \partial_x^2 J) \right\} d\tau \right]. \quad (2.11)$$

Now, notice that the right-hand side of the PDE (2.8a) can be expressed as the minimum value of a quadratic form in u_s as follows:

$$-\partial_\tau J = -\frac{1}{2} (g_\tau^\top \partial_x J + S_\tau^\top)^\top R_\tau^{-1} (g_\tau^\top \partial_x J + S_\tau^\top) + V_\tau + f_\tau^\top \partial_x J + \frac{1}{2} \text{Tr} (\sigma_\tau \sigma_\tau^\top \partial_x^2 J) \quad (2.12a)$$

$$= \min_{u_\tau} \left[\frac{1}{2} u_\tau^\top R_\tau u_\tau + S_\tau u_\tau + V_\tau + (f_\tau + g_\tau u_\tau)^\top \partial_x J + \frac{1}{2} \text{Tr} (\sigma_\tau \sigma_\tau^\top \partial_x^2 J) \right]. \quad (2.12b)$$

Therefore, for an arbitrary choice of u_τ , we have

$$-\partial_\tau J \leq \frac{1}{2} u_\tau^\top R_\tau u_\tau + S_\tau u_\tau + V_\tau + (f_\tau + g_\tau u_\tau)^\top \partial_x J + \frac{1}{2} \text{Tr} (\sigma_\tau \sigma_\tau^\top \partial_x^2 J) \quad (2.13)$$

where the equality holds iff

$$u_\tau = -R_\tau^{-1} (g_\tau^\top \partial_x J + S_\tau^\top). \quad (2.14)$$

Combining (2.11) and (2.13), we obtain

$$\begin{aligned} J(x, t) &\leq \mathbb{E}^{x,t} \psi(X_{t_f}) + \mathbb{E}^{x,t} \left[\int_t^{t_f} \left\{ \frac{1}{2} u_\tau^\top R_\tau u_\tau + S_\tau u_\tau + V_\tau(X_\tau) \right\} d\tau \right] \\ &= C(x, t, u_{[t, \infty)}). \end{aligned} \quad (2.15)$$

This proves the statement (i). Since (2.15) holds with equality if and only if (2.14) is satisfied, statement (ii) also follows. $\blacksquare$

Theorem 2.1 shows that solving the HJB equation (2.8) is equivalent to solving the stochastic optimal control problem (2.3). However, since the HJB PDE rarely admits analytical solutions, numerical methods must be employed. Direct discretization is feasible only for low-dimensional problems (e.g., $n \leq 3$).

Exercise 2.1 (Continuous-Time Stochastic Linear Quadratic Regulator) Consider the special case where the state equation (2.1) is linear:

$$dX_t = AX_t + Bu_t(X_t) + FdW_t, \quad X_{t_0} = x_0 \in \mathbb{R}^n, \quad t \in [t_0, t_f], \quad (2.16)$$

where A , B , and F are matrices of appropriate dimensions, and t_f is a fixed terminal time. Assume the cost function (2.2) is quadratic:

$$V_t(X_t) = \frac{1}{2}X_t^\top Q_t X_t, \quad \psi(X_{t_f}) = \frac{1}{2}X_{t_f}^\top Q_{t_f} X_{t_f}, \quad (2.17)$$

where Q_t and Q_{t_f} are positive semidefinite matrices. Show that the HJB PDE (2.8) reduces to the continuous-time Riccati equation. Derive the optimal state feedback law by simplifying (2.9).

Exercise 2.2 (Feedback Cooling) Recall the Langevin equation for a particle with mass m in the heat bath of temperature T derived in Exercise 1.3. For notational consistency, denote by X_t the velocity of the particle, serving as the state of the system. If the particle is subject to the external force u_t , the Langevin equation is modified to

$$dX_t = \left(-\frac{\gamma}{m}X_t + \frac{1}{m}u_t \right) dt + \frac{\sqrt{2\gamma kT}}{m}dW_t, \quad X_{t_0} = x_0 \quad (2.18)$$

where W_t is the standard Brownian motion. Assuming that u_t is the control input, consider the state feedback policy to minimize the cost functional

$$\int_{t_0}^{t_f} \mathbb{E} \left[X_t u_t + \frac{1}{2} R u_t^2 \right] dt$$

where $R > 0$. There is no terminal cost, i.e., $\psi(X_{t_f}) = 0$.

(i) For each $x \in \mathbb{R}$ and $t \in [t_0, t_f]$, define the value function by

$$J(x, t) = \min_{u[t, t_f]} \int_t^{t_f} \mathbb{E} \left[X_\tau u_\tau + \frac{1}{2} R u_\tau^2 \right] d\tau.$$

Derive the HJB equation that $J(x, t)$ satisfies.

(ii) Assume that $J(x, t) = \frac{1}{2}p_t x^2 + q_t$ where p_t and q_t are time-varying parameters. Derive the ODEs for p_t and q_t . Express the optimal state feedback law using p_t .

(iii) Compute the limit $p^* = \lim_{t \rightarrow -\infty} p_t$. What is the corresponding time-invariant state feedback law? Assuming that t_f is fixed, compute

$$J^* := \lim_{t_0 \rightarrow -\infty} \frac{1}{t_f - t_0} \int_{t_0}^{t_f} \mathbb{E} \left[X_t u_t + \frac{1}{2} R u_t^2 \right] dt.$$

Hint: The general solution to the scalar Riccati ordinary differential equation $\dot{y}_t = ay_t^2 + by_t + c$ with $a \neq 0$ and $\Delta = b^2 - 4ac > 0$ is given by

$$y_t = \frac{1}{2a} \left(-b + \sqrt{\Delta} \frac{k - e^{t\sqrt{\Delta}}}{k + e^{t\sqrt{\Delta}}} \right)$$

where $k \in \mathbb{R}$ is a free parameter.

(iv) Notice that the quantity

$$\lim_{t_0 \rightarrow -\infty} \frac{1}{t_f - t_0} \int_{t_0}^{t_f} \mathbb{E} [X_t u_t] dt$$

represents the average work done by the controller to the particle. If this quantity is negative, it means that the controller extracts work from the particle submerged to the heat bath. By taking the limit $R \searrow 0$, compute the maximum rate at which the controller can extract work.

(i) Substituting $f_t = -\frac{\gamma}{m}x$, $g_t = \frac{1}{m}$, $\sigma_t = -\frac{\sqrt{2\gamma kT}}{m}$, $V_t(x) = 0$, $R_t(x) = R$, and $S_t(x) = x$ into (2.8a), we obtain

$$-\partial_t J = -\frac{1}{2R} \left(\frac{1}{m} \partial_x J + x \right)^2 - \frac{\gamma x}{m} \partial_x J + \frac{\gamma kT}{m^2} \partial_x^2 J.$$

The boundary condition is $J(x, t_f) = 0 \forall x \in \mathbb{R}$.

(ii) We obtain

$$\begin{aligned} -\dot{p}_t &= -\frac{1}{R} \left(\frac{1}{m} p_t + 1 \right)^2 - \frac{2\gamma}{m} p_t \\ -\dot{q}_t &= \frac{\gamma kT}{m} p_t. \end{aligned}$$

The optimal state feedback control law is $u_t = -\frac{1}{m} \left(\frac{p_t}{m} + 1 \right) x_t$.

(iii) We obtain $p^* = -m(1 + \gamma R) + m\sqrt{(1 + \gamma R)^2 - 1}$. The corresponding time-invariant state feedback law is $u_t = \left(\gamma - \frac{1}{R} \sqrt{(1 + \gamma R)^2 - 1} \right) x_t$. Since $\dot{q}_t = -\frac{\gamma kT}{m^2} p_t$ and $q_{t_f} = 0$, we obtain $q_{t_0} = \frac{\gamma kT}{m^2} \int_{t_0}^{t_f} p_t dt$. This implies that $\lim_{t_0 \rightarrow -\infty} \frac{1}{t_f - t_0} q_{t_0} = \frac{\gamma kT}{m^2} p^*$. It follows that

$$J^* = \frac{\gamma kT}{m^2} p^* = \frac{\gamma kT}{m} \left(-1 - \gamma R + \sqrt{(1 + \gamma R)^2 - 1} \right).$$

(iv) $J^* \rightarrow -\frac{\gamma kT}{m}$ as $R \searrow 0$. Hence, $\frac{\gamma kT}{m}$ is the maximum possible rate of extraction of work.

2.3 Control of Partially Observable Systems

The optimal control law derived in Theorem 2.1 is a state-feedback policy in the sense that the optimal policy (2.9) is a (generally nonlinear) function of X_t . However, the problem formulation (2.3) itself does not require the policy to be state-feedback; it only requires that u_t be an $\mathcal{F}_t$ -adapted process. Theorem 2.1 thus *proves* that state-feedback is optimal among all admissible control laws.

In many practical scenarios, state-feedback is not an admissible policy. Specifically, consider the following partially observable system:

$$dX_t = (f_t(X_t) + g_t(X_t)u_t) dt + \sigma_t(X_t)dW_t, \quad X_{t_0} = x_0 \in \mathbb{R}^n, \quad (2.19a)$$

$$dY_t = h_t(X_t)dt + dV_t, \quad (2.19b)$$

where V_t is a multi-dimensional standard Brownian motion independent of W_t . Assume that the controller observes Y_t but not X_t .

To impose this constraint formally, let $\mathcal{F}_t^Y$ be the increasing family of σ -algebras generated by Y_t . Clearly, $\mathcal{F}_t^Y \subset \mathcal{F}_t$ for all t , implying that $\mathcal{F}_t^Y$ contains less information than $\mathcal{F}_t$. The optimal control problem of interest is given by

$$\min_{u_{[t_0, \infty)}} C(x_0, t_0, u_{[t_0, \infty)}) \quad (2.20a)$$

$$\text{s.t.} \quad \text{Dynamics (2.19),} \quad (2.20b)$$

$$u_t \text{ is } \mathcal{F}_t^Y \text{-adapted.} \quad (2.20c)$$

The admissible control u_t is required to satisfy all the conditions imposed in the state feedback case (2.3). Additionally, u_t is subject to (2.20c), which formally indicates that (2.20) is a partially observable control problem.

Recall that a partially observable Markov decision process (POMDP) can be reformulated as a fully observable MDP using a belief-state representation.¹ The same idea applies to (2.20). Let π_t denote the probability density function (assuming it exists) representing the posterior distribution of X_t conditioned on the observation history Y_s , $0 \leq s \leq t$. One can view π_t as a fully observable *belief state* whose evolution is governed by a stochastic partial differential equation (SPDE). It can be shown that the optimal control u_t solving (2.20) is a deterministic function of π_t .

2.4 Nonlinear Filtering

For simplicity, let $u_t = 0$ and consider the following partially observable system:

$$dX_t = f_t(X_t)dt + \sigma_t(X_t)dW_t, \quad X_{t_0} \sim \pi_0, \quad (2.21a)$$

$$dY_t = h_t(X_t)dt + dV_t. \quad (2.21b)$$

We focus on the *nonlinear filtering problem*, i.e., the problem of computing π_t based on the observation history Y_s , $0 \leq s \leq t$. Here, we only state the main results without derivations. Full proofs are beyond the scope of this lecture.

The following SPDE, known as the *Kushner equation* [2], describes the belief-state dynamics:

$$d\pi_t = - \sum_i \frac{\partial}{\partial x^i} (f_t^i \pi_t) dt + \frac{1}{2} \sum_{ijk} \frac{\partial^2}{\partial x^i \partial x^j} (\sigma_t^{ik} \sigma_t^{jk} \pi_t) dt + \pi_t (h_t - \mathbb{E}^{\pi_t}[h_t])^\top (dY_t - \mathbb{E}^{\pi_t}[h_t] dt). \quad (2.22)$$

This is an SPDE interpreted in the sense of Ito. Note that the functions $\pi_t(x)$, $f_t(x)$, $\sigma_t(x)$, and $h_t(x)$ depend on x (although the dependence on x is dropped for notational simplicity), and the equation (2.22) holds pointwise for each $x \in \mathbb{R}^n$. In (2.22), the i -th entry of the vector f_t is denoted by f_t^i , whereas the (i, j) entry of the matrix σ_t is denoted by σ_t^{ij} . Since $\mathbb{E}^{\pi_t} h_t = \int h_t(x) d\pi_t(x)$ is linear in π_t , the Kushner equation (2.22) is a nonlinear SPDE in terms of π_t .

¹However, such a transformation does not necessarily make the problem computationally tractable, since the belief space is typically much higher-dimensional than the original state space.

The following SPDE, known as the *Zakai equation* [3], describes the dynamics of the *unnormlized*² belief state p_t :

$$dp_t = - \sum_i \frac{\partial}{\partial x^i} (f_t^i p_t) dt + \frac{1}{2} \sum_{ijk} \frac{\partial^2}{\partial x^i \partial x^j} (\sigma_t^{ik} \sigma_t^{jk} p_t) dt + p_t h_t^\top dY_t. \quad (2.23)$$

Unlike the Kushner equation, the Zakai equation is linear in p_t .

Exercise 2.3 (Kalman-Bucy Filter) Consider the following linear, partially observable system:

$$dX_t = A_t X_t dt + F_t dW_t, \quad X_{t_0} \sim \mathcal{N}(x_0, P_0), \quad (2.24a)$$

$$dY_t = C_t X_t dt + dV_t, \quad (2.24b)$$

where A_t , F_t , and C_t are time-varying matrices of appropriate dimensions. Show that the Kushner equation (2.22) reduces to the Kalman-Bucy filter:

$$d\hat{X}_t = A_t \hat{X}_t dt + P_t C_t^\top (dY_t - C_t \hat{X}_t dt), \quad \hat{X}_{t_0} = x_0, \quad (2.25a)$$

$$\frac{d}{dt} P_t = A_t P_t + P_t A_t^\top + F_t F_t^\top - P_t C_t^\top C_t P_t, \quad P_{t_0} = P_0. \quad (2.25b)$$

²That is, $p_t \propto \pi_t$, but $\int p_t(x) dx \neq 1$. The normalized belief can be recovered by $\pi_t(x) = p_t(x) / \int p_t(x) dx$.

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